

CAREERMATRI · CAREER DECISION BRIEF

Respondent

Mechanical Engineering · PhD · PRO · 2026-10-05

DECISION

The three to look at first

1

Mechanical Design & Product Engineering

Interest High

Experience High

Core 0/6 confirmed

Build evidence

2

CAE / Simulation / Computational Engineering

Interest High

Experience High

Core 0/9 confirmed

Build evidence

3

Thermal / Fluids / Energy Systems

Interest High

Experience High

Core 0/6 confirmed

Build evidence**What matters most in this result**

- The experience you recorded currently reaches Organisational value is confirmed.
- Mechanical Design & Product Engineering and CAE / Simulation / Computational Engineering sit close together, so this is the point to compare the two.

Your most important evidence gap**Interpreting requirements**

Mechanical Design & Product Engineering · Turned the given conditions into values you could work with, such as dimensions, loads and performance

What to do next

Interpreting requirements do one piece of work that fills that gap. Turned the given conditions into values you could work with, such as dimensions, loads and performance

Nothing here measures your ability or your odds of being hired. We wrote only what your answers and recorded experience confirm.

DECISION

Comparing the roles to look at first

Where the cells diverge is what tells you what to do next.

Role	Work style	Interest	Experience	Learning intent	Core confirmed	Evidence depth	Current
Mechanical Design & Product Engineering	Further away	High	High	High	0 / 6	E4 Organisational value	Build evid
CAE / Simulation / Computational Engineering	Further away	High	High	High	0 / 9	E4 Organisational value	Build evid
Thermal / Fluids / Energy Systems	Further away	High	High	High	0 / 6	—	Build evid
Worth keeping in view							
Automation / Robotics / Mechatronics	Further away	High	Low	Low	0 / 6	—	Look at th
Engineering Data / Digital Engineering	Further away	High	Medium	Low	0 / 6	E4 Organisational value	Look at th

We do not merge the cells into one. Where interest is high but experience is empty, doing one small thing first beats applying, and comes first. To say that, the two cells have to be separated. All sixteen roles are in the appendix.

WHY

how what you learned is used in real work

This runs in one line from what you learned to what the organisation counts as a result.

Mechanical Design & Product Engineering

Settling on one buildable form while the requirements pull against each other

What you learned

Mechanics of materials Statics

This is what you met in class. Having taken a course is not the same as being able to do it

Real work

turn the requirements into dimensions and tolerances

set the structure with the loads in mind

compare the design options and choose one

Technical judgement

Thickness and geometry

Material

Safety margin

Output

Design options and drawings that go on to production

3D models and drawings

Design calculation sheet

What the organisation values

Performance

cost

manufacturability

Private company is the basis.

My evidence today

bracket weight reduction capstone · Organisational value

Evidence still needed

A drawing or 3D model taken through to completion

A comparison table of design options

A record of building it and correcting it

EVIDENCE

The evidence I have today

This looks at how far one experience has travelled, from activity through to repeatability. A lower rung does not mean something is lacking. It is the range confirmed by what you have recorded.

bracket weight reduction capstone

Activity	Confirmed	turned the requirements into dimensions and compared three design options
Judgement	Confirmed	checked the rib layout and thickness that cost less to machine · rib layout
Output	Confirmed	drawing · analysis report
Performance	Confirmed	5% margin against the allowable stress in test · compared against the allowable stress
Organisational value	Confirmed	adopted as the final design
Repeatability	Not yet	

Repeatability to get there Have you used the same method on another project

ANSYS Mechanical

Activity	Confirmed	ANSYS Mechanical you have used
Judgement	Not yet	
Output	Not yet	
Performance	Not yet	
Organisational value	Not yet	
Repeatability	Not yet	

Judgement to get there What did you analyse

SolidWorks · Design and CAD

through to an output

Experience of use	Confirmed
Connected experience	bracket weight reduction capstone
where it was used	compare geometry options
what you judged	rib layout
what remains	drawing
what it was compared against	compared against the allowable stress

ANSYS Mechanical · CAE and structural analysis

Experience of use

Experience of use	Confirmed
Connected experience	Not connected

A tool name on its own only confirms the activity. count as a score.

ORGANIZATION VALUE

The same technical background creates different value in different organisations

The same knowledge produces a product in one place and a paper in another.

Private company

Output

Design options and drawings that go on to production

What counts as performance

Performance · cost · manufacturability · lead time

Government-funded research institute

Output

Design as a prototype and as a project deliverable

What counts as performance

Project goals met · technical level · transferability

University laboratory

Output

A prototype that shows the concept, and the reasoning behind its design

What counts as performance

Novelty · reproducibility · scholarly contribution

Engineering and technical consultancy

Output

A comparison of design options the client uses to decide

What counts as performance

Client requirements met · schedule · repeat business

The same Mechanical Design & Product Engineering knowledge reads as a different result in each organisation. Your fit does not change when the organisation does.

bracket weight reduction capstone

picked the rib layout and thickness that cost less to machine, drawing · analysis report left behind, 15% margin against the allowable stress in test compared against

What an interview will ask Have you used the same method on another project

Application · interview · portfolio

Application

Write one line about what you decided rather than what you did. The sentence above You can carry it across as it stands.

Interview

Mechanical Design & Product Engineering asks what your judgement rested on: why you chose it and Prepare what you gave up alongside it.

Portfolio

One page of conditions reads before a photo of the result.

WHY · EVIDENCE

Mechanical Design & Product Engineering

Mechanical Design & Product Engineering

Build evidence

Settling on one buildable form while the requirements pull against each other

I want to work on weighing competing requirements and deciding the structure and shape of a product or a part · I want to go deeper into design, drawings, tolerances and product development · I am fine with work that means long stretches at a computer with models, drawings and data is the side you answered toward.

WHAT THE WORK ACTUALLY IS

turn the requirements into dimensions and tolerances

set the structure with the loads in mind

compare the design options and choose one

take it through design review

JUDGEMENTS MADE OFTEN

Thickness and geometry

Material

Safety margin

Tolerance allocation

LINK TO ACADEMIC KNOWLEDGE

Mechanics of materials

Statics

WHAT THE ORGANISATION SEES AS A RESULT

Performance

cost

manufacturability

lead time

MY EVIDENCE TODAY

From your current input, the core evidence for this role is not yet is not sufficiently confirmed.

EVIDENCE STILL NEEDED

not yet

Interpreting requirements

Turned the given conditions into values you could work with, such as dimensions, loads and performance

not yet

Generating and comparing design options

Set a criterion across two or more options and chose one

not yet

Judging geometry, dimensions and material

Set thickness, geometry and material on the basis of load and material

Next action Interpreting requirements Do one piece of work that leaves a record.

WHY · EVIDENCE

CAE / Simulation / Computational Engineering

CAE / Simulation / Computational Engineering

Build evidence

Screening design options by calculation before anything is built

I want to work on verifying by calculation or simulation whether a design holds up under real load, temperature and flow conditions · I want to go deeper into analysis, simulation and numerical computation · I am fine with work that means long stretches at a computer with models, drawings and data is the side you answered toward.

WHAT THE WORK ACTUALLY IS

build the analysis model

set the boundary conditions and loads

compare the design options

reconcile it with the test values

JUDGEMENTS MADE OFTEN

Elements and mesh

Material model

Boundary conditions

Convergence criteria

LINK TO ACADEMIC KNOWLEDGE

Mechanics of materials

WHAT THE ORGANISATION SEES AS A RESULT

Design criteria met

agreement with test

shorter development time

Whether the standard was met

MY EVIDENCE TODAY

From your current input, the core evidence for this role is not yet is not sufficiently confirmed.

EVIDENCE STILL NEEDED

not yet

Problem definition

Framed,
yourself,
what had to
be analysed
and why

not yet

Model creation

Simplified
the
geometry
and settled
the
elements
and the
mesh

not yet

Assumptions set out

Wrote down
the
assumptions
you solved
under

Next action Problem definition Do one piece of work that leaves a record.

WHY · EVIDENCE

Thermal / Fluids / Energy Systems

Thermal / Fluids / Energy Systems

Build evidence

Designing the path heat and fluid take, so that temperature and pressure come out right

I want to work on calculating or measuring heat, fluid and energy flows to improve how a system performs is the side you answered toward.

WHAT THE WORK ACTUALLY IS

calculate the thermal and flow loads

design the cooling and the system layout

set the operating conditions

validate the performance

JUDGEMENTS MADE OFTEN

Cooling method

Flow and temperature conditions

Equipment capacity

System layout

LINK TO ACADEMIC KNOWLEDGE

no course has matched yet

WHAT THE ORGANISATION SEES AS A RESULT

Temperature

reliability

power consumption

cost

MY EVIDENCE TODAY

From your current input, the core evidence for this role is not yet is not sufficiently confirmed.

EVIDENCE STILL NEEDED

not yet

Load calculation

Calculated
how much
heat or flow
was needed

not yet

System layout

Settled the
layout for
how it
would flow

not yet

Selecting equipment and conditions

Set the
capacity
and the
operating
conditions

Next action Load calculation Do one piece of work that leaves a record.

GAP

Not yet confirmed

This does not mean you cannot do it. It means that from what you have recorded, it is not confirmed.

Experience not yet translated into performance evidence

In these experiences, the cells an organisation reads as a result are still empty.

What you recorded **ANSYS Mechanical**

Confirmed so far Activity

Not yet confirmed

- What did you analyse
- Which conditions did you vary as you compared
- What did you decide on the back of that result

This does not say you lack experience. It means there are cells you lived through but have not yet written down.

Use \u2018Fill in your experience\u2019 in the web report and you can complete these After you fill them in and regenerate the report, this section changes.

Just pick from the options

What you pick here is not part of your fit. It only changes how far your evidence is confirmed.

Interpreting requirements · What judgement was that result used for

Whether the result led to a decision is what applications and interviews read.

What judgement was the result used for you may pick more than one

Design option selection Dimensional change Material change Condition or setting change
Deciding the test conditions Root cause analysis Decided whether to continue or stop
It was not used directly in a decision

Who made that decision

I decided I proposed it and we decided together Someone else decided It was already fixed

Generating and comparing design options · What did you weigh it against when choosing

It reads as your own decision only when there is a standard you compared against.

What did you compare the result against you may pick more than one

Design criteria Experimental values An existing product An alternative design Theoretical values
Another analysis method Codes and standards I did not compare

On what basis did you choose you may pick more than one

Performance Weight Cost Difficulty of manufacture Schedule Safety margin Reliability
It was fixed, so there was nothing to choose Something else

How far each role's core areas are confirmed

Role	Confirmed so far	Partially confirmed	Not yet confirmed	Fill this first
Mechanical Design & Product Engineering	0 / 6	0	6	Interpreting requirements · Generating and comparing design options
CAE / Simulation / Computational Engineering	0 / 9	0	9	Problem definition · Model creation
Thermal / Fluids / Energy Systems	0 / 6	0	6	Load calculation · System layout

ACTION

The next experience to build

Do not start several things. Fill the one gap that is widest right now.

What is still empty

Interpreting requirements

Turned the given conditions into values you could work with, such as dimensions, loads and performance

- 1 something small you can scope: Mechanical Design & Product Engineering pick one problem on that side
- 2 write down the conditions and assumptions first
- 3 Interpreting requirements do the equivalent yourself
- 4 Performance · cost check it against
- 5 where it differed, write down why and correct it
- 6 leave it as a two-page record

What to leave behind Design options and drawings that go on to production · 3D models and drawings · why you chose it · what you compared it against · its limits

What to do, and when

30 days

- **Interpreting requirements** fill that one gap and nothing else. Turned the given conditions into values you could work with, such as dimensions, loads and performance
- write down, project by project, what you decided and what you were handed

This plan comes from the gaps you have right now. Record more experience and this line changes.

90 days

- **Interpreting requirements · Generating and comparing design options** carry it through to an output and a comparison against a standard
- take one method and show it works on another problem
- set two roles side by side and see which reads your evidence better

This plan comes from the gaps you have right now. Record more experience and this line changes.

365 days

- produce material that translates your research depth into the organisation's language of outputs
- write the same experience up two ways, one for each organisation type you are aiming at
- use the same method on another problem so that repeatability is on record

This plan comes from the gaps you have right now. Record more experience and this line changes.

APPENDIX

These were left out of the main report. Use them when you want to trace something back.

All sixteen role families

The main report carries only the three to look at first and the two to keep in view.

Role	Work style	Interest	Experience	Learning intent	Core confirmed	Current stage
Mechanical Design & Product Engineering	Further away	High	High	High	0 / 6	Build evidence
CAE / Simulation / Computational Engineering	Further away	High	High	High	0 / 9	Build evidence
Thermal / Fluids / Energy Systems	Further away	High	High	High	0 / 6	Build evidence
Automation / Robotics / Mechatronics	Further away	High	Low	Low	0 / 6	Look at this first
Engineering Data / Digital Engineering	Further away	High	Medium	Low	0 / 6	Look at this first
R&D / Advanced Development	Further away	High	Low	Low	0 / 8	Look at this first
Research Scientist / Academic Research	Further away	Medium	Medium	Low	0 / 7	Look at this first
Systems Engineering / Technical PM	Further away	Medium	Low	—	0 / 6	Look at this first
Materials / Tribology / Mechanical Failure	Further away	Medium	Low	—	0 / 5	Look at this first
Technical Consulting / Engineering Services	Further away	Medium	Low	—	0 / 6	Look at this first
Manufacturing / Production Engineering	Further away	Low	Low	Low	0 / 6	Set aside for now
Process / Industrialization Engineering	Further away	Low	Low	Low	0 / 6	Set aside for now
Quality / Reliability / Failure Analysis	Further away	Low	Low	Low	0 / 6	Set aside for now
Equipment / Maintenance / Facilities	Further away	Low	Low	Low	0 / 5	Set aside for now
Test / Verification / Validation	Further away	Low	Low	Low	0 / 6	Set aside for now
Applications / Field / Customer Engineering	Further away	Low	Low	—	0 / 5	Set aside for now

which questions produced this line

The main report drops the question numbers and gives the reason in plain words. When you want to trace it back we leave them here for you.

Role	Questions behind it
Mechanical Design & Product Engineering	Q1, Q17, Q41, Q50, Q53
CAE / Simulation / Computational Engineering	Q2, Q18, Q42, Q50
Thermal / Fluids / Energy Systems	Q10
Automation / Robotics / Mechatronics	Q9, Q24, Q45
Engineering Data / Digital Engineering	Q15, Q22, Q46, Q50

Work style in detail

This axis rests on two questions, so we put no number on it. Until it is validated we describe it only as close, mixed or further away.

Axis	Leaning	Questions
Independent ↔ Collaborative	Neither side	2
Challenge ↔ Stability	Neither side	2
Speed ↔ Quality	Neither side	2

whether a result produced once can be reused

produced a result once **Confirmed** jacket weight reduction capstone

used the same method again **Not yet**

carried it to another problem **Not yet**

handed it over for others to use **Not yet**

A degree does not raise it. It rises only from the projects and methods you recorded.

Full evidence coverage by role

Core, supporting and optional are all laid out. An empty optional area is never held against you.

Mechanical Design & Product Engineering

Core areas

Confirmed 0

Partial 0

Not yet 6

all 6

Confirmed so far

- ✓ Use of design tools Optional
bracket weight reduction capstone · Tools: SolidWorks

Not yet confirmed

- ☐ Interpreting requirements
- ☐ Generating and comparing design options
- ☐ Judging geometry, dimensions and material
- ☐ CAD and drawing outputs
- ☐ Structural and functional review
- ☐ Validation or design review
- ☐ Tolerances Supporting
- ☐ Manufacturability Supporting
- ☐ Cost Supporting
- ☐ Change control Supporting
- ☐ Interfaces Supporting
- ☐ Feedback from real build and test Supporting
- ☐ Industry standard Optional

CAE / Simulation / Computational Engineering

Core areas

Confirmed 0

Partial 0

Not yet 9

all 9

Confirmed so far

- ✓ Convergence check Supporting
bracket weight reduction capstone
- ✓ Use of analysis tools Optional
ANSYS Mechanical · Tools: ANSYS Mechanical

Not yet confirmed

- ☐ Problem definition
- ☐ Model creation
- ☐ Assumptions set out
- ☐ Boundary conditions and loads
- ☐ Material and model parameters
- ☐ Running the analysis
- ☐ Interpreting the result
- ☐ Validation
- ☐ Link to a design decision
- ☐ Mesh independence Supporting
- ☐ Parametric study Supporting
- ☐ Automation and scripting Supporting
- ☐ Documenting the analysis Supporting
- ☐ Large-scale computation Optional
- ☐ Optimisation Optional

Thermal / Fluids / Energy Systems**Core areas**

Confirmed 0

Partial 0

Not yet 6

all 6

Not yet confirmed

- ☐ Load calculation
- ☐ System layout
- ☐ Selecting equipment and conditions
- ☐ Analysis and calculation results
- ☐ Performance check
- ☐ Reflected in the design
- ☐ Efficiency and energy Supporting
- ☐ Transient behaviour Supporting
- ☐ Operational control Supporting
- ☐ Test rig Supporting
- ☐ Renewables and power generation Optional
- ☐ HVAC design Optional

Automation / Robotics / Mechatronics**Core areas**

Confirmed 0

Partial 0

Not yet 6

all 6

Not yet confirmed

- ☐ Deciding the scope of automation
- ☐ Mechanism and cell design
- ☐ Actuator and sensor selection
- ☐ Control implementation
- ☐ Commissioning
- ☐ Automation result
- ☐ Safety design Supporting
- ☐ Vision and inspection Supporting
- ☐ Offline validation Supporting
- ☐ Return on investment Supporting
- ☐ A specific robot or PLC Optional
- ☐ Material handling automation Optional

Engineering Data / Digital Engineering**Core areas**

Confirmed 0

Partial 0

Not yet 6

all 6

Confirmed so far

- ✓ Programming language Optional
bracket weight reduction capstone

Not yet confirmed

- ☐ Choosing what to automate
- ☐ Data structure design
- ☐ Building tools and scripts
- ☐ Result validation
- ☐ How it was used
- ☐ Someone else used it
- ☐ Usage instructions Supporting
- ☐ Change history Supporting
- ☐ Connecting to an existing system Supporting
- ☐ Visualisation Supporting
- ☐ Predictive model Optional

We do not turn the number of confirmed cells into a single We do not merge them into a number. Merging hides what the other cells were. An empty optional area is never held against you.

how the material was produced

- Interest, experience, decision ownership, work style and learning intent come from different questions and are read separately. We never merge them into one score.
- Experience is not part of your fit. It only changes how far your evidence is confirmed, differ.
- The evidence ladder is how deep one experience goes; evidence coverage is how much of the role's territory it covers. We never merge the two.
- A degree does not raise your evidence rung.

Limits

- Cognitive interviews and the pilot are still ahead, so we do not report a measurement error.
- These are not measures of ability or of your odds of being hired.
- The analysis below is written against the role itself. We did not ask for a target country, so Country-specific material such as licensing, salary, hiring practice and work authorisation is not included. Written without verified data, someone will believe it and act on it.
- The sixteen role families and their evidence areas hold for mechanical engineering only.

Version

Assessment	ME_V2_DECISION_2026
Result schema	3.0
Evidence map	1.0
Questions	92 of 92 answers
Generated	2026-10-05